

Collaborative Differentially Private Federated Learning Framework for the Prediction of Diabetic Retinopathy

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Abstract—The diagnosis of diabetic retinopathy may be streamlined and expedited with the help of deep learning, which is an efficient way to help an eye specialist examine the enormous amount of retinal images. For these strategies to be effective, big datasets must be consolidated and used for training. Medical data privacy laws frequently make it impossible to gather and share patient data on a single system. In this paper, we introduce a collaborative differentially private federated learning system that enables deep learning image analysis without transferring patient data between healthcare organizations. We investigated four different machine learning algorithms—AlexNet, ResNet50, SqueezeNet1.1, and VGG16—for varying amounts of noise using a dataset of 35120 retina images divided into five classes—No Diabetic Retinopathy, Mild, Moderate, Severe, and Proliferative Diabetic Retinopathy (PDR). Our ResNet50 model outperformed the state-of-the-art diabetic retinopathy prediction models with an accuracy 83.05% when we added no noise, and with an accuracy 79.35% with a noise multiplier of 8.0. By including our checkpoint techniques, we have reduced the total communication overhead by 49% when compared to federated learning without checkpoints.

I. INTRODUCTION

Patients without immediate access to physicians benefit from deep neural networks (DNNs) that are capable of recognizing disease from medical data. Research on healthcare patient data like [1] using deep learning creates a new milestone. Medical images are processed with DNNs for faster diagnosis of skin disease [2], lung cancer [3], or diabetic retinopathy (DR) [4]. The memorization capacity of DNNs, however, can be ill-treated by adversaries to regenerate a patient's data [5] or extrapolate a patient's participation in the training dataset [6], [7]. Medical data rarely can be found in a centralized dataset due to privacy risks and legal restrictions; for this reason, it has become increasingly popular to train on distributed datasets that preserve privacy and utility [8].

A federated learning concept has evolved in recent years to address privacy concerns about sharing potentially sensitive patient information among healthcare systems. In [9], [10] researchers introduced federated learning (FL)-based architecture to detect contraband in x-ray baggage security images while maintaining user privacy. Federated learning can be adopted to address the challenges of using real-world clinical data. Research on federated learning is going up, as [11] is

research on self-regulated federated learning. Without ethical or legitimate constraints, a decentralized prediction engine could potentially use data from independent health systems. A decentralized privacy-controlled federated learning architecture was implemented in [12] to predict the complications of three diseases caused by diabetes, using federated datasets from different healthcare systems without sharing data. [13] utilized federated learning (FL) to train the machine learning model where hospitals shared local datasets with a server which is centralized. During each communication round, the server sends the current model to each hospital, which updates it on their private datasets and sends it back to the server. Due to the overfitting of models, hospitals' updates can leak patient information [5].

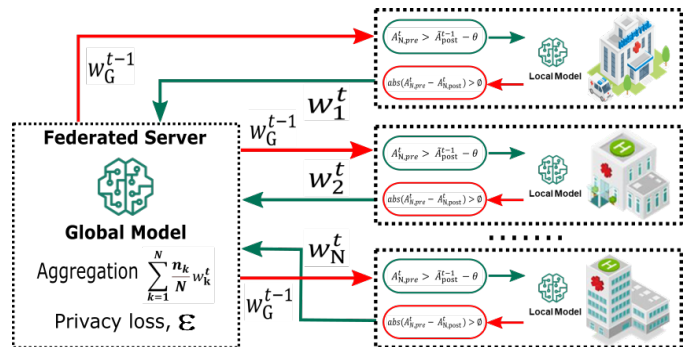


Fig. 1: Federated Learning on decentralized healthcare systems: 1) A global model is initialized by FL server, 2) Global model state is sent to participating nodes (hospitals), 3) Nodes train the local model with their individual dataset and send the model updates back to the server, 4) The server aggregates and processes these local updates into a single global update and applies it to the global model.

Differential Privacy is a randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ with a domain $\mathcal{D}$ (e.g., possible training datasets) and range $\mathcal{R}$ (e.g., all possible trained models) that satisfies (ϵ, δ) -differential privacy if for any two adjacent datasets $d, d' \in \mathcal{D}$ and for any subset of outputs $\mathcal{S} \subseteq \mathcal{R}$ it holds that $Pr[\mathcal{M}(d) \in \mathcal{S}] \leq e^\epsilon Pr[\mathcal{M}(d') \in \mathcal{S}] + \delta$, where the probability space is over the coin flips of the mechanism $\mathcal{M}$. If $\delta = 0$, we say that $\mathcal{M}$ is ϵ -differentially private.

We present a collaborative differentially private federated

learning framework for the prediction of diabetic retinopathy that enables deep learning image analysis without sharing patient data between healthcare organizations. Using a dataset of 35120 retina images classified into five classes—No Diabetic Retinopathy, Mild, Moderate, Severe, and Proliferative Diabetic Retinopathy. These five classes are represented with five real retinal images in figure 2. We examined the performance of four different machine learning algorithms for different levels of noise: AlexNet, ResNet50, SqueezeNet1.1, and VGG16 (PDR). Our ResNet50 model outperformed the most complex models for predicting diabetic retinopathy with accuracy of 83.05% and 79.35%, respectively, when we added no noise and a noise factor of 8.0. Our checkpoint techniques have decreased the total communication overhead by 49% compared to FL without checkpoints.

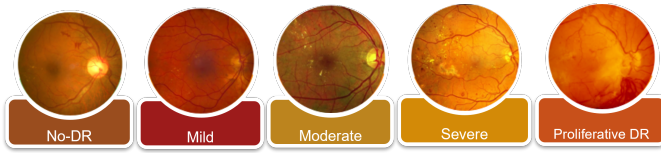


Fig. 2: Real images of five different classes from the diabetic retinopathy dataset.

II. SYSTEM MODEL

Consider that three hospitals either in three separate countries or in the same country in different states wish to work together to develop a global model for classifying diabetic retinopathy. However, none of them are eager to share their own patient's data because of privacy issues. An ML model is shared between edge nodes and the central server to describe how data is preserved. The architectural flow shown in Figure 1 can help us comprehend the mechanism's operation better. The coordinator in this scenario is the central aggregation server, often referred to as the FL server. A global model is initialized by the FL server, and the global model state is sent to participating nodes 1-3 (hospitals). Each of the local data owners 1-3 trains the model with their own dataset by applying the previous round's global update and keeping its gradient differentially private by adding a certain level of noise, and then transmitting the current round's local update to the server for secure aggregation. The server aggregates and processes these local updates into a single global update and applies it to the global model. After integrating the model updates collected from the data owners, it gives the combined model updates back to the local nodes. This approach is repeated until the model converges or the specified number of iterations have been carried out. The local data owners' raw data is never removed from this architecture. With this approach, user privacy and data security are ensured while reducing communication overhead. By not exchanging real patient's retinal images across different hospitals or healthcare, we can guarantee data privacy by utilizing the Differentially private federated learning method. Instead, we distribute the training model from various edge nodes to the global server, which may then push it to the edge nodes.

III. RELATED WORK

Recent years have seen a great deal of research in federated learning on medical data. As federated learning allows collaborative learning without sharing private data, in a medical context this learning process provides sensitive data privacy. When working with federated learning (FL) in a medical setting, data privacy is just one of several variables that must be taken into account. Computational and communication efficiency are also important considerations. Real-world huge clinical data are being used for predicting several health complications as well as drug discovery. Some studies concentrate primarily on efficiency, while others concentrate mostly on privacy.

By preventing clients from taking part in model updates when a client had irrelevant data, Talukder et al. [14] work to reduce resource waste in federated learning by saving communication round. Asad et al. [15] introduced clients' sampling probability where they used $P=10\%$ is called the participation ratio, meaning each iteration is 10 percent of the total base number of participants. Luping et al. [16] propose Communication-Mitigating Federated Learning (CMFL), which reduces communication costs by uploading only relevant local updates. FedMMD is proposed in [17] as a method for addressing the communication issue for non-IID datasets, where participants learn from each other to fix the loss function by incorporating MMD. Caldas et al. proposed in [18] to increase the server-to-participant communication efficiency by using lossy compression and federated dropout (Fed-Dropout).

In recent years, there was an increased surge of contribution in the field of preserving privacy in federated learning. One of the research papers authored by Xue et al. [19] provides an in-depth explanation of different possible attacks that may threaten the differential model. There are multiple ways these attacks can be categorized such as sensitive data recovery, data poisoning, obtained model stealing, and trapdoor attacks. To deal with some of these attacks few approaches were proposed in recent years like the Preserving privacy model. Chang et al. [20] talked about some of these privacy-related issues in differential learning. Primarily authors divided attacks into two categories, one during the training process and another during the prediction process. Zhang et al. [21] talks about Privacy preservation to deal with some of these attacks. In [22], researchers focused more on applying differential privacy techniques in model gradients, they consider each hospital means client holds non-malicious and non-colluding data. Differentially Private stochastic gradient descent (DPSGD) is used in [22]. Two techniques like Selective parameter update and Sparse vector are adopted by [23] to provide strong protection against indirect data leakage. An individual client can share only a limited amount of information by using selective parameter-sharing methods.

IV. METHODOLOGY

Our contributions include a) identifying the important research area, and the training process on decentralized data from hospital IOT devices; b) introducing an algorithm that can be applied to this environment for improving communication

and computational efficiency. More particularly, we offer the Federated Averaging approach, which is a computer that does model averaging, by taking the differentially private Local stochastic gradient descent (SGD) for each client. We run thorough testing on this approach to demonstrate that it is robust against balance for IID data distributions and can reduce the number of communication links needed to train sophisticated deep networks on decentralized data.

Algorithm 1 DPSGD training with FedSRC

Input: K : number of hospitals, $\mathbb{D}$: dataset, g : sgd, ω : model's trainable parameters, $\mathcal{L}(\cdot, \cdot)$: loss function, q : sampling probability, σ : noise scale, C : gradient norm bound, η : learning rate, β : momentum, T : number of rounds, (ϵ, δ) : bounds on record-level DP loss, $\mathbb{A}$: model accuracy, θ : client's regulating parameter, ϕ : global update restriction parameter

Output: ω_G^0 : optimized global model.

$\bar{\epsilon} = 0$

$[\bar{\epsilon}, \hat{K}] \leftarrow \text{EstimateEpsilonOverClients}(\epsilon, K)$:

for $t \in [1, 2, \dots, T]$ **do**

for $k \in [1, 2, \dots, \hat{K}]$ **do**

$\mathbb{D}_k^t = \text{Sampling}(\mathbb{D}_k)$

$\mathbb{A}_{k,pre}^t = \text{CalculateAccuracy}(\mathbb{D}_k^t, \omega_G^{t-1})$

if $\mathbb{A}_{k,pre}^t > (\tilde{\mathbb{A}}_{post}^{t-1} - \theta)$ **then**

for $x_i \in \mathbb{D}_k^t$ **do**

$g^t(x_i) = \Delta \mathcal{L}(\omega_G^{t-1}, x_i)$

$\bar{g}^t(x_i) = g^t(x_i) / \max(1, \frac{\|g^t(x_i)\|_2}{C})$

end

$\tilde{g}_k^t = \frac{1}{|\mathbb{D}_k^t|} (\sum_{x_i \in \mathbb{D}_k^t} \bar{g}^t(x_i) + \mathcal{N}(0, \frac{\sigma^2 \cdot C^2 \cdot I}{K}))$

$g_k'^t = \tilde{g}_k^t + \beta g_k'^{t-1}$

$\omega_k^t = \omega_G^{t-1} - \eta g_k'^t$

$\mathbb{A}_{k,post}^t = \text{CalculateAccuracy}(\mathbb{D}_k^t, \omega_k^t)$

if $abs(\mathbb{A}_{k,pre}^t - \mathbb{A}_{k,post}^t) > \phi$ **then**

$\omega_k^t = 0$

end

end

end

$\tilde{\mathbb{A}}_{post}^t = \text{Median}_{k \in K^t}(\mathbb{A}_{k,post}^t)$

$\bar{\epsilon} = \text{GetPrivacySpent}(\delta, q, \sigma, t)$

if $\bar{\epsilon} > \bar{\epsilon}$ **then**

 return ω_G^{t-1}

end

$\omega_G^t = \frac{1}{K} (\text{SecureAggregation}(\sum_k \omega_k^t))$

 Broadcast(ω_G^t)

end

In this study, Algorithm 1 is described the training procedure, where federated SGD is performed among K hospitals. During each communication round t, each hospital trains a batch of samples from its own dataset, $\mathbb{D}_k^t \subset \mathbb{D}_k$, where each sample is picked independently and with sampling probability q which determines whether a training point is likely to be included in a mini-batch. Batch size is not specified due to independent sampling, and it's an arbitrary variable with parameters q and

$|\mathbb{D}_k|$. Let $|\mathbb{D}_k|$ denote the batch size and η for the learning rate. Here, C indicates the maximum value (it gives control that limits the optimizer's sensitivity to individual training points) of the L2-norm (Euclidean norm) of per-sample gradients. If there is a greater norm per-sample gradient than this, then its norm is clipped to C [24]. Since hospitals distrust the server, there is a probable solution for each hospital to add a large-scale noise for model updating, ω_k^t to ensure it is differentially private. However, a large amount of noise addition for the purpose of model updating has the unwanted effect of declining the model performance means accuracy. So, finding the trade-off between accuracy and privacy budget to secure aggregation of model updates would be a better solution, preventing the server from discovering hospital model updates. Since the global server is now unable to see the local model update, less amount of noise can be added to the model's weight (noise is added to stochastic gradient descent) to make their model updates differentially private. In Algorithm 1, secure aggregation after calculating differentially private stochastic gradient descent is introduced in [22]. During each client's model training Gaussian noise $\mathcal{N}(0, \frac{\sigma^2 \cdot C^2 \cdot I}{K})$ is added with the gradient value. After generating differentially private stochastic gradient descent, we need to calculate the privacy loss with respect to δ , noise scale σ , sampling probability q , and t . We then compare this privacy loss to the threshold privacy loss, $\bar{\epsilon}$. Here, the threshold value is estimated over the number of clients per round by applying Algorithm 2. Now, If the calculated loss is greater than the threshold value, we don't need to continue the next round with the current round's global update that indicates the privacy budget doesn't provide the privacy guarantee and will not provide it in next rounds also, so at this moment, it is obvious that previous round's global model updated, ω_G^{t-1} is optimized. Other than that, we continue secure aggregation and broadcast the global model update to the clients.

After sampling $\mathbb{D}_k^t$ is used for obtaining model accuracy, $\mathbb{A}_{k,pre}^t$ from the global update model, ω_G^{t-1} . In Algorithm 1 checkpoint-1 is set after sampling, where $\mathbb{A}_{k,pre}^t$ is pre-trained model accuracy is compared with the previous round's median post-trained Accuracy $\tilde{\mathbb{A}}_{post}^{t-1} - \theta$, where θ is a parameter that controls how aggressively clients regulate themselves. checkpoint-1 will determine whether a client is allowed for training its local model or not. Meanwhile, checkpoint 2 is responsible for determining whether the global server will receive the client's model update for aggregation or not. Before this checking, the local model's post-trained accuracy, $\mathbb{A}_{k,post}^t$ is obtained with respect to $\mathbb{D}_k^t$ and ω_k^t . Here, the absolute difference of $\mathbb{A}_{k,pre}^t$ and $\mathbb{A}_{k,post}^t$ is being compared with ϕ is a parameter that controls how restrictive checkpoint-2 is in client update regulation. If the absolute difference is higher than ϕ that means if the client sends its model update to global then this will lead to global modal diverging. In Algorithm 1, ω_k^t is set to zero to mean that the client's t th round model's update will not be sent to the global server. At each round, a client's model update that is allowed to pass checkpoint 2 will add to the global aggregating process to obtain the global

model update. In this study, in order to get privacy spent from the four parameters (δ, q, σ, t) , (α, ϵ) -RDP, alternatively, it is, ϵ -Renyi differential privacy of order α [25], a natural relaxation of differential privacy mechanism is applied. Here, $M(\cdot)$ in definition (1) is a (α, ϵ) -RDP mechanism that satisfies (ϵ', δ) -DP, (where $\epsilon' = \epsilon + \frac{\log(1/\delta)}{\alpha-1}$) for any $0 < \delta < 1$.

Privacy accounting is a procedure that can be either executed offline, for example, before the training part, so that we can know in advance how long it will take to train a model given some DP constraints, or online and stop the training phase once an upper limit for ϵ or δ has been reached. There are mainly three variables to consider when it comes to training federated learning with DP constraints; 1) The number of steps, T , that the Gaussian Mechanism is applied to the model, 2) the number of expected clients, K , per round we would need to get acceptable privacy and 3) Noise multiplier - the ratio of the standard deviation, σ (noise scale), of the Gaussian noise to the L_2 -sensitivity of the function to which the noise is added. The threshold value of privacy loss, $\bar{\epsilon}$ will be used to determine the number of clients we can take per round, and the noise level we can use for DPSGD to get optimal accuracy without exceeding $\bar{\epsilon}$. What privacy loss (epsilon, ϵ), do we obtain if we scale up the noise multiplier by this number of clients per round? To answer this, we introduced Algorithm 2, where the binary search is used for determining the clients per round and the value of noise multiplier which will later be used for model training for each global round and will estimate privacy budget, ϵ . In order to get ϵ over clients per round (α, ϵ) -RDP for a list of finite α is applied. Here, the privacy budget is checked with a given value, ϵ with respect to a number of clients. At the final stage, $O(\log(n))$ search technique is applied to achieve an optimal privacy budget and corresponding clients which are two required parameters for each global round.

Algorithm 2 Threshold ϵ estimation over clients per round

Input: K : base clients per round, ϵ : target epsilon

Output: $\hat{\epsilon}$: threshold epsilon, $\hat{K}$: clients per round.

EstimateEpsilonOverClients(ϵ, K) :

$\hat{\epsilon} \leftarrow \text{getEpsilon}(K)$

if $\hat{\epsilon} < \epsilon$ **then**

return $[\hat{\epsilon}, K]$

end

$\bar{K} \leftarrow K$

// Increase $\bar{K}$ exponentially until target epsilon

do

$\bar{K} \leftarrow 2 * \bar{K}$

$\hat{\epsilon} \leftarrow \text{getEpsilon}(\bar{K})$

while ($\hat{\epsilon} \geq \epsilon$);

$[\hat{\epsilon}, \hat{K}] \leftarrow \text{SEARCH}(\bar{K}, \epsilon)$

return $[\hat{\epsilon}, \hat{K}]$

V. EXPERIMENTAL RESULTS

1) *Dataset*: We have experimented dataset from the Kaggle repository which is a resized version of the Diabetic Retinopathy Kaggle competition dataset contains 35126 images (left and

right eye images of each) of 17563 persons. In this competition, the problem is to classify the images of patients' retinas into five categories: (No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR). In order to recompense for the distinction in image dimensions, all images were resized to the same dimensions of 224×224. We consider IID data partitions (i.e, we shuffle the dataset). From this labeled dataset we use 28101 retinal images for our experiment for training and 7025 (20% of the entire dataset) for evaluation (testing).

Here, one of our observations regarding datasets is that a well-distributed dataset is the most important part when working with deep learning. The accuracy of the model depends to a large extent on the dataset. The data set we experiment with in our research is not balanced, or well distributed. We have another copy of the dataset from Kaggle, which is almost the same in terms of data distribution. As the health care data is not public always, we train the model just by using the Kaggle dataset, but the accuracy for the test dataset is always at a constant level (0.71-0.74) for five different classes. We perform all kinds of hyperparameter tuning during model training, but there is no improvement in the accuracy level for five classes. However, to increase the accuracy, first, we considered 19316 images to get a balanced dataset for model training. Figure 3 shows the model performance for two different sets of output classes (a) for five classes and (b) for two classes, and the maximum accuracy is 74.09% (when the number of output classes is 5) and 83.05% (for binary classes), when there is no noise that means not differentially private. Mainly we thereafter define DR and No-DR meaning only two classes. We consider all class labels greater than 0 as class label 1 and others as 0. In this case, we observe a variation in accuracy, and that is more than 83%. By this, we are pretty sure that balanced data plays an important role in order to get more accuracy.

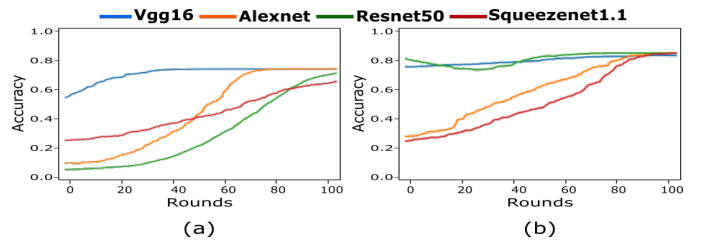


Fig. 3: (a) Comparison of model performance for five classes (No-DR, Mild, Moderate, Severe, Proliferative DR) (b) Comparison of model performance for two classes (No-DR, DR).

A. Experimental Data

1) *Check-pointing*: We have used the checkpointing technique in differential private federated learning as a way to reduce the overall communication rounds. Checkpointing is a way to compare two different accuracies of the same local client and determine if any communication update is required to be sent to the global server from the client node. The accuracy is calculated after training the local model with Retina images and further, we compare these metrics with old accuracy obtained from the same client with old input data. We mainly have three different contexts where we can witness accuracy change in

local clients. One is to introduce bad data as an input, the second is to perform different cross-validation and the third is to add more retina images as input sources. Out of the above three approaches we specifically included as bad quality and new data introduction for accuracy checks for checkpoints. We have started our experiment with 400 clients and a fixed portion clients of the total are randomly selected for each round, we would evaluate the output accuracy for local models. If we see any accuracy change we would send an update to the global server else we would ignore selected clients for the next rounds. Using the check-pointing technique helped reduce overall global and client communication to almost 49 percent. This helped us reduce compute resources and network bandwidth.

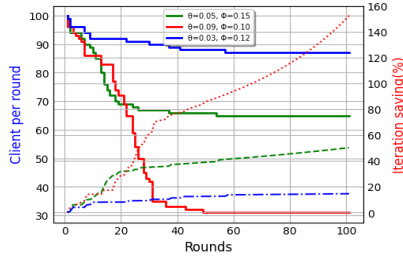


Fig. 4: Iteration saving per round based on the number of clients per round.

We can observe from Figure 4 that communication rounds decrease as we iterate through the process. After each iteration, we would evaluate the output weights to that of existing previous weights and determine whether to send the client update to the global server or not.

2) *Model quality and Privacy trade-off*: As we apply Differential privacy in Federated learning few things we need to consider always are the client number, the value ϵ , and the noise multiplier. Adding noise with SGD will decline the performance of the model, but the amount of noise can be controlled in an average update at each round with two knobs: the standard deviation of the Gaussian noise added to the sum, and the number of clients in the average. We experiment with a relatively small number of clients per round to determine how much noise the model can tolerate with an acceptable loss of model performance. Then to train the final model, we can increase the amount of noise in the sum, while proportionally scaling up the number of clients per round (this way many clients can be supported per round due to the size of the dataset). Initially, we train the model by adding increasing amounts of noise to 56 clients per round. This is done by adjusting the noise multiplier, which is a ratio of noise standard deviation to the clipping norm. This model seems to be able to tolerate a noise multiplier of 0.56 with 56 expected clients per round without deteriorating its performance. There seems to be a degradation in the model when the noise multiplier is 0.9, and 1.2. We know there is a general trade-off between privacy and modal quality. The more noise we use, the more privacy we will get for the same number of rounds and a number of clients. Alternatively, less noise doesn't necessarily lead to more privacy but more accuracy we may get. In order to achieve our target

privacy level for the round, we must determine the number of clients. The ϵ values are estimated based on δ , noise-multiplier, and the number of clients per round by adopting [26], [27] and with the help of the TensorFlow differential privacy function.

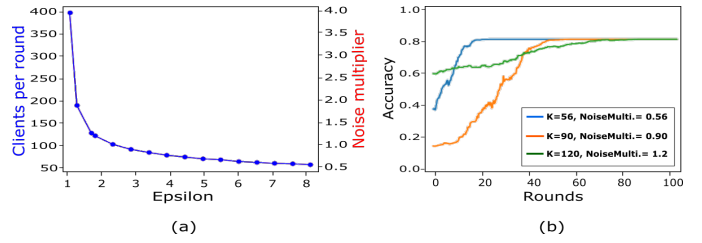


Fig. 5: (a) Estimated epsilon value over client (b) Model performance with a different number of clients per round and noise multiplier.

In Figure 5, we show that the relationship between the epsilon, noise-multiplier and client per round indicates which epsilon and noise-multiplier value is appropriate for what number of clients per round. Our estimation is done by implementing Algorithm 2 and the right figure describes the model accuracy with the given number of clients per round and the value of the noise multiplier. We experiment by training ResNet50 for clients 56, 90, and 90 per round with the values of noise-multiplier 0.56, 0.9, and 1.2 respectively. In this way, this estimation technique is useful to figure out the minimum number of expected clients per round that would allow us to maintain privacy.

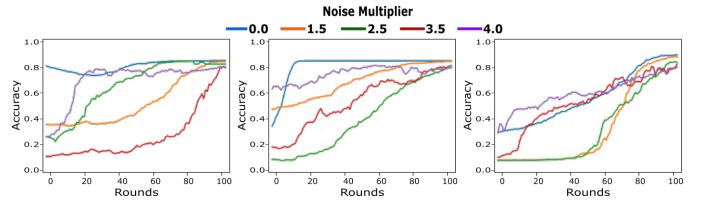


Fig. 6: Accuracy of four different models at five different noise multipliers.

3) *Models performance in Federated Learning for DR data*: We use SqueezeNet1.1 [28]: a comparatively lighter convolutional neural network provides. In this study, we work on four different ML models, which are widely used for image classification. As we work on federated learning model sharing between clients and global servers, the size of the model is a significant factor for both training and testing, where SqueezeNet1.1 can be used to leverage better communication efficiency and for our case using this ML model and applying checkpoints, we were able to save 49 percent communication overhead. In addition to training SqueezeNet1.1, we also train AlexNet, VGG16, and ResNet50. But the accuracy we get is the best performance from ResNet50, and the accuracy is 83.05%.

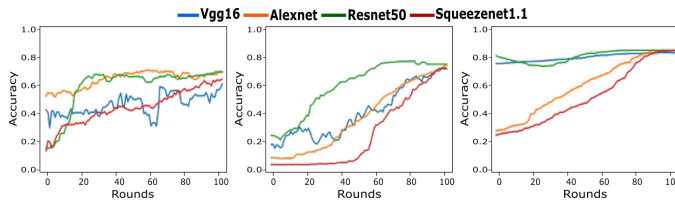


Fig. 7: Accuracy of four different models at noise multipliers - high noise (8.0), medium noise (3.5), and no noise (0).

As presented in Figure 6, applying different values of noise level and a certain magnitude of δ ($1e-6$) and as well as per rounds clients number, we show the accuracy for three machine learning models (first SqueezeNet1.1, second ResNet50, and third for VGG16), and it shows ResNet50 does the comparatively better performance than others. This result shows in this figure as noise is increasing the model accuracy degrades. As we are aware of the trade-off between privacy and model quality, Figure 7 shows the four different models' performance upon three different values of noise-multiplier. It is evident that accuracy decreases as noise increases. This figure shows the accuracy gain of the models for three different levels: high noise (noise multiplier = 8.0), medium noise (noise multiplier = 3.5), and no noise (noise multiplier = 0).

VI. CONCLUSIONS

In this paper, we provide an architecture based on differentially private federated learning (FL) to predict diabetic retinopathy while protecting patients' privacy. Our differentially private ML model is trained and tested utilizing SqueezeNet1.1, AlexNet, and ResNet50 opening the way for large-scale autonomous classification of diabetic retinopathy throughout the world through collaboration. Depending on the expectation, accuracy, and category type, we can adjust the input parameters and obtain prime results while preserving data security. Even though our experiment's performance is comparable to the benchmark performance of non-FL algorithms, it highlights the difficulties and promise of using FL to classify diabetic retinopathy from individual healthcare patients' retinal images.

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